

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 10/01/2026**
Time: 3 Hrs **GTM - 14** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. If $\hat{e}_1$ and $\hat{e}_2$ are two unit vectors and θ is the angle between them, then $\sin\left(\frac{\theta}{2}\right)$ is

(A) $\frac{1}{2}|\hat{e}_1+\hat{e}_2|$ (B) $\frac{1}{2}|\hat{e}_1-\hat{e}_2|$ (C) $\frac{\hat{e}_1.\hat{e}_2}{2}$ (D) $\frac{|\hat{e}_1\times\hat{e}_2|}{2}$

2. Consider the equation $\frac{d}{dt}\left(\int F.ds\right)=A(F.p)$, where F = force, s = displacement, t = time and p = momentum. The dimensional formula of A will be

(A) $[M^0L^0T^0]$ (B) $[ML^0T^0]$ (C) $[M^{-1}L^0T^0]$ (D) $[M^0L^0T^{-1}]$

3. A car starting from rest, accelerates at the rate f through a distance s, then continues at constant speed for time t and then decelerates at the rate of $\frac{f}{2}$ to comes to rest. If the total distance traversed is 15s, then

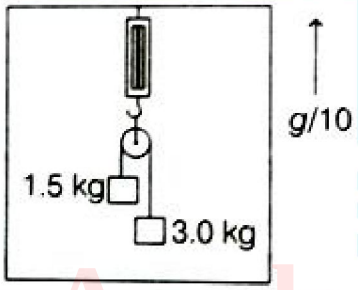
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(A) $s = \frac{1}{36}ft^2$ (B) $s = \frac{1}{18}ft^2$ (C) $s = \frac{1}{4}ft^2$ (D) $s = \frac{1}{72}ft^2$

4. A battalion of soldiers is ordered to swim across a river 500 ft wide. At what minimum rate should they swim perpendicular to river flow in order to avoid being washed away by the waterfall 300 ft downstream. The speed of current is 3 mph.

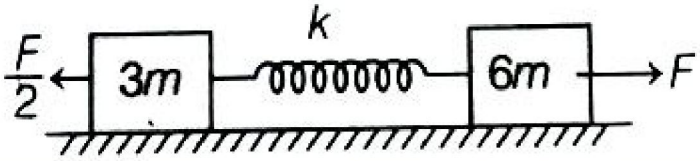
(A) 6 mph (B) 5 mph (C) 4 mph (D) 2 mph

5. Find the reading of the spring balance as shown in figure. The elevator is going up with an acceleration of $\frac{g}{10}$, the pulley and the string are light and the pulley is smooth.



(A) 4 kg (B) 4.3 kg (C) 4.4 kg (D) 4.5 kg

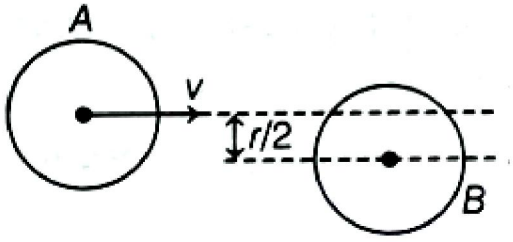
6. For shown situation find the maximum elongation in the spring. Neglect friction everywhere. Initially, the blocks are at rest and spring is unstretched.



(A) $\frac{4F}{3k}$ (B) $\frac{3F}{4k}$ (C) $\frac{4F}{k}$ (D) $\frac{2F}{k}$

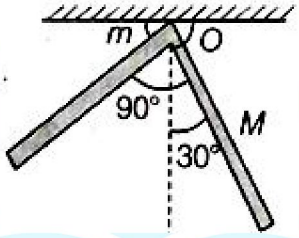
7. A disk A of radius r moving on perfectly smooth surface at a speed v undergoes an elastic collision with an identical stationary disk B. Find the velocity of the disk B after collision if the impact parameter is r/2 shown in the figure.

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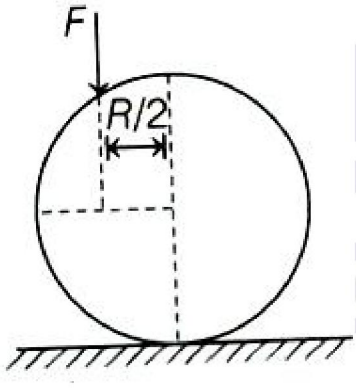
(A) $\frac{v\sqrt{15}}{4}$ (B) $\frac{v}{4}$ (C) $\frac{v}{2}$ (D) $\frac{\sqrt{3}v}{2}$

8. Two uniform rods of equal length but different masses are rigidly joined to form an L-shaped body, which is then pivoted about O as shown. If in equilibrium the body is in the shown configuration, ratio M/m will be



(A) 2 (B) 3 (C) $\sqrt{2}$ (D) $\sqrt{3}$

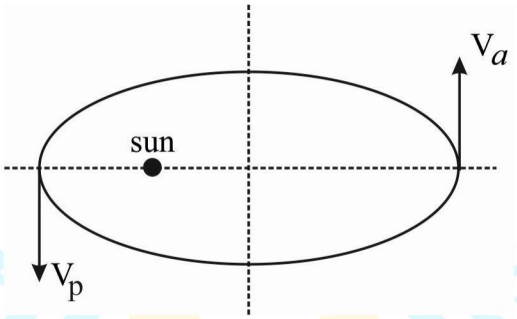
9. How much is the magnitude of the frictional force acting at the point of contact for a cylinder which is rolling (without slipping) under the action of an external force F as shown in the diagram below? (Assume F to be constant and applied constantly at the same point on the cylinder as shown)



(A) $\frac{F}{3}$ (B) $\frac{2F}{3}$ (C) $\frac{F}{2}$ (D) $\frac{3F}{2}$

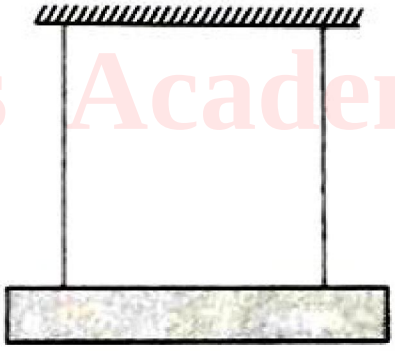
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10. A planet moves around Sun in an elliptical orbit of eccentricity e. The ratio of the velocity at perigee v_p and at apogee v_a is given by



(A) $\frac{V_p}{V_a}=\frac{1+e}{1-e}$ (B) $\frac{V_p}{V_a}=\frac{1-e}{1+e}$ (C) $\frac{V_p}{V_a}=\sqrt{\frac{1+e}{1-e}}$ (D) $\frac{V_p}{V_a}=\sqrt{\frac{1-e}{1+e}}$

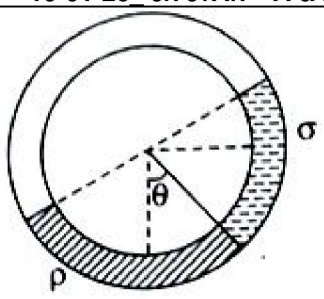
11. Two wires of equal length and cross-section area are suspended as shown in figure. Their Young's modulus are Y_1 and Y_2, respectively. The equivalent Young's modulus will be



(A) Y_1+Y_2 (B) $\frac{Y_1+Y_2}{2}$ (C) $\frac{Y_1Y_2}{Y_1+Y_2}$ (D) $\sqrt{Y_1Y_2}$

12. A small uniform tube is bent into a circle of radius r whose plane is vertical. The equal volumes of two fluids whose densities are ρ and $\sigma(\rho>\sigma)$, fill half the circle. Find the angle that the radius passing through the interface makes with the vertical.

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(A) $\cos\theta = \frac{\rho-\sigma}{\rho+\sigma}$ (B) $\tan\theta = \frac{\rho-\sigma}{\rho+\sigma}$ (C) $\sin\theta = \frac{\rho+\sigma}{\rho-\sigma}$ (D) $\sin\theta = \frac{\rho}{\sigma}$

13. In the experiment for measurement of viscosity 'η' of given liquid with a ball having radius R, consider following statements.

a) Graph between terminal velocity V and R will be a parabola

b) The terminal velocities of different diameter balls are constant for a given liquid.

c) Measurement of terminal velocity is dependent on the temperature.

d) This experiment can be utilized to assess the density of a given liquid.

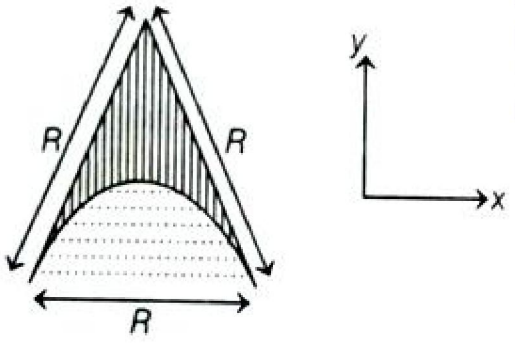
e) If balls are dropped with some initial speed, the value of η will change

Choose the correct answer from the options given below:

A) b, d and e only B) a, c and d only

C) c, d and e only D) a, b and e only

14. A light, rigid sheet of triangular shape has a curved portion cut from it as shown in figure. It floats on the surface of water. Some soap solution is dropped over dotted region. Surface tension of water and soap film are T_1 and T_2, respectively. $T_1 = 1.5T_2$. Mark correct option.



(A) The frame experiences a net force $F = (T_1 - T_2)R$ in $y < 0$ direction

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(B) The frame experiences force $F=(T_1-T_2)R$ in $y>0$ direction

(C) The frame experiences force $(T_1-T_2)(2-\pi)R$ in $y>0$ direction

(D) Resultant force on wire frame is zero

15. The diameter of a rod is given by $d=d_0(1+ax)$, where a is a constant and x is the distance from one end. If the thermal conductivity of the material is K, what is the thermal resistance of the rod if length is l.

(A) $\frac{1}{K\pi d_0^2}$ (B) $\frac{4\ell}{K\pi d_0^2(a\ell+1)}$

(C) $\frac{2\ell}{K\pi d_0^2(a\ell+1)^2}$ (D) $\frac{2\ell}{K\pi d_0^2(a\ell+1)}$

16. In solar radiation, the intensity of radiation is maximum around the wavelength λ. If R is the radius of the sun and c is the velocity of light, the mass lost by the sun per seconds is proportional to

(A) $\frac{R^2}{\lambda^4c^2}$ (B) $\frac{R^2}{\lambda^2c^2}$ (C) $\frac{R^3}{\lambda^4c^3}$ (D) $\frac{R^3}{\lambda^4c^2}$

17. One mole of an ideal gas undergoes a process $p=p_0\left[1+\left(\frac{2V_0}{V}\right)^2\right]^{-1}$, where p_0, V_0 are constants. Change in temperature of the gas when volume is changed from $V=V_0$ to $V=2V_0$ is

(A) $\frac{4}{5}\frac{p_0V_0}{nR}$ (B) $\frac{3}{4}\frac{p_0V_0}{nR}$ (C) $\frac{2}{3}\frac{p_0V_0}{nR}$ (D) $\frac{7}{9}\frac{p_0V_0}{nR}$

18. A polytropic process for an ideal gas is represented by equation $pV^n=$ constant. If γ is ratio of specific heats $\left(\frac{C_p}{C_v}\right)$, then value of n for which molar heat capacity of the process is negative, is given as

(A) $\gamma>n$ (B) $\gamma>n>1$

(C) $n>\gamma$ (D) None, as it is not possible

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19. A point mass is subjected to two simultaneous sinusoidal displacements in x-direction, $x_1(t) = A\sin\omega t$ and $x_2(t) = A\sin\left(\omega t + \frac{2\pi}{3}\right)$. Adding a third sinusoidal displacement $x_3(t) = B\sin(\omega t + \phi)$ brings the mass to a complete rest. The values of B and ϕ are

(A) $\sqrt{2}A, \frac{3\pi}{4}$ (B) $A, \frac{4\pi}{3}$ (C) $\sqrt{3}A, \frac{5\pi}{6}$ (D) $A, \frac{\pi}{3}$

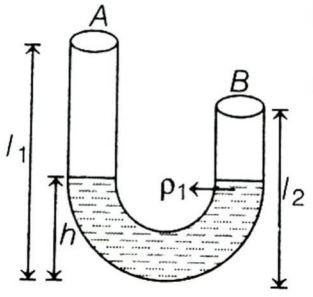
20. At $t = 0$, a transverse wave pulse travelling in the +ve x-direction with a speed of $2m/s$ in a wire is described by $y = \frac{6}{x^2}$, given that $x \neq 0$. Transverse velocity of a particle at $x = 2m$ and $t = 2s$ is

(A) $3m/s$ (B) $-3m/s$ (C) $8m/s$ (D) $-8m/s$

SECTION – II
(Numerical Value Answer Type)

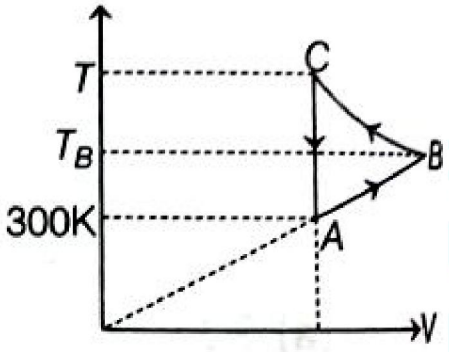
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A U-tube having uniform cross-section but unequal arm length $\ell_1 = 100$ cm and $\ell_2 = 50$ cm has same liquid of density ρ_1 filled in it upto a height $h = 30$ cm as shown in figure. Another liquid of density $\rho_2 = \frac{\rho_1}{2}$ is poured in arm A. Both liquids are immiscible. What length of the second liquid (in cm) should be poured in A, so that second overtone of A is in unison with fundamental tone of B? (Neglect end correction)

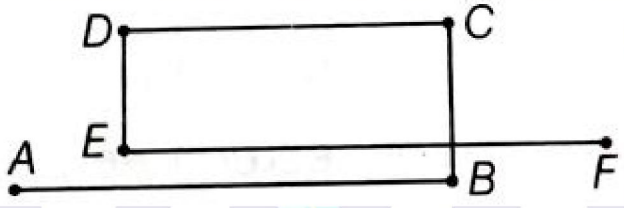


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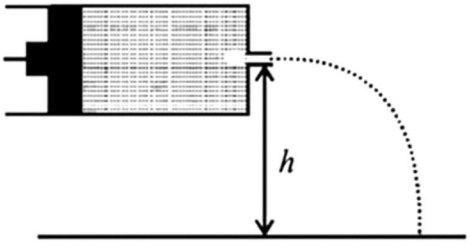
22. Consider the cyclic process ABCA shown in figure, performed on a sample of 1 mole of an ideal gas. A total of 1000 J of heat is withdrawn from the sample in the process. The work done on the gas during part BC is 1830 J. Find the value of T_B (in K). $(R = 8.3\ J/K-mol)$



23. The length of sections AB and EF is ℓ_1 in the figure, and their coefficient of thermal expansion is same $\alpha_1 = 2\times10^{-5}/°C$. The length of section CD is ℓ_2 and its expansion coefficient is $\alpha_2 = 1\times10^{-5}/°C$. What must be the ratio $\frac{\ell_2}{\ell_1}$ to ensure that distance AF remains the same at all temperature?
(Points A, B, E and F reside on the same line)

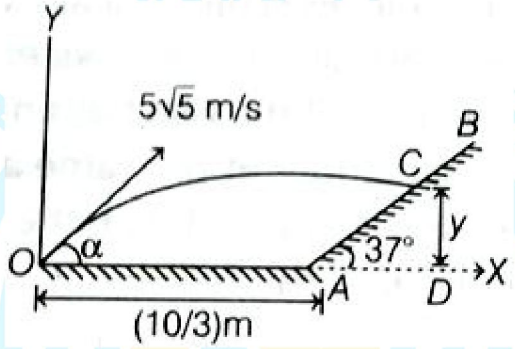


24. Suppose a tube has two area of cross-sections as shown in figure in which radii of its both ends are 2 mm and 0.5 mm. If piston is moving with constant velocity of 0.25 m/s then range of water falling on horizontal surface(in m) is; $\left(h = 5m, g = 10m/s^2\right)$



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25. A particle is projected from point O on the ground with velocity $u = 5\sqrt{5}$ m/s at angle $\alpha = \tan^{-1}(0.5)$. It strikes at a point C on a fixed smooth plane AB having inclination of $37°$ with horizontal as shown in figure. Angle made by velocity vector with horizontal (X-axis) is $\theta°$ at C then find θ. $(g = 10 m/s^2)$



CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. $XNa_2HAsO_3 + YNaBrO_3 + ZHCl \rightarrow NaBr + H_3AsO_4 + NaCl$

The values of X, Y and Z in the above redox reaction are respectively :

(A) 2, 1, 2 (B) 2, 1, 3 (C) 3, 1, 6 (D) 3, 1, 4

27. The wave number of the first emission line in the Balmer series of H-Spectrum is :

(R = Rydberg constant) :

(A) $\frac{5}{36}R$ (B) $\frac{9}{400}R$ (C) $\frac{7}{6}R$ (D) $\frac{3}{4}R$

28. Trigonal bipyramidal shape is shown by :

(A) $XeOF_2$ (B) XeO_3F_2 (C) $FXeOSO_2F$ (D) $[XeF_8]^{2-}$

29. The order of increasing sizes of atomic radii among the elements O, S, Se and As is :

(A) As < S < O < Se (B) Se < S < As < O

(C) O < S < As < Se (D) O < S < Se < As

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30. In an atom how many orbital (s) will have the quantum numbers; $n = 3, \ell = 2$ and $m_\ell = +2$?

(A) 5 (B) 3 (C) 1 (D) 7

31. Given :

(a) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell);\ \Delta H^\circ_{298K} = -285.9\ kJ\ mol^{-1}$

(b) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g);\ \Delta H^\circ_{298} = -241.8\ kJ\ mol^{-1}$

The molar enthalpy of vaporization of water will be :

(A) $241.8\ kJ\ mol^{-1}$ (B) $22.0\ kJ\ mol^{-1}$

(C) $44.1\ kJ\ mol^{-1}$ (D) $527.7\ kJ\ mol^{-1}$

32. Which one of the following molecules is polar?

(A) XeF_4 (B) IF_5 (C) SbF_5 (D) CF_4

33. In which of the following ionization processes the bond energy has increased and also the magnetic behavior has changed from paramagnetic to diamagnetic?

(A) $NO \rightarrow NO^+$ (B) $N_2 \rightarrow N_2^+$ (C) $C_2 \rightarrow C_2^+$ (D) $O_2 \rightarrow O_2^+$

34. Electron gain enthalpy with negative sign of fluorine is less than that of chlorine due to :

(A) High ionization enthalpy of fluorine

(B) Smaller size of chlorine atom

(C) Small size of fluorine atom

(D) Bigger size of 2p orbital of fluorine

35. Solid $Ba(NO_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At which concentration of Ba^{2+}, precipitate of $BaCO_3$ begins to form? (K_{sp} of $BaCO_3 = 5.1\times10^{-9}$)

(A) 5.1×10^{-5} M (B) 7.1×10^{-8} M (C) 4.1×10^{-5} M (D) 8.1×10^{-7} M

36. Photons of an electromagnetic radiation has an energy 11 keV each. To which region of electromagnetic spectrum does it belong?

(A) X-ray region (B) Ultra violet region

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(C) Infrared region (D) Visible region

37. An ideal gas at atmospheric pressure is adiabatically compressed so that its density becomes 32 times of its initial value. If the final pressure of gas is 128 atmospheres, the value of 'γ' of the gas is :

(A) 1.5 (B) 1.4 (C) 1.3 (D) 1.6

38. Given that 1g of water in liquid phase has volume 1 cm^3 and in vapour phase 1671 cm^3 at atmospheric pressure and the latent heat of vaporization of water is 2256 J / g ; the change in the internal energy in Joules for 1 g of water at 373 K when it changes from liquid phase to vapour phase at the same temperature is :

(A) 2256 (B) 167 (C) 2089 (D) 1

39. But-2-yne is reacted separately with one mole of Hydrogen as shown below :

$$\underline{B} \xleftarrow[\text{liq } NH_3]{Na} CH_3 - C \equiv C - CH_3 + H_2 \xrightarrow[\Delta]{Pd/BaSO_4/S/\text{Quinoline}} A$$

a. A is more soluble than B.

b. The boiling point & melting point of A are higher and lower than B respectively.

c. A is more polar than B because dipole moment of A is zero.

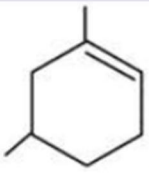
d. Br_2 adds easily to B than A

Identify the incorrect statements from the options given below:

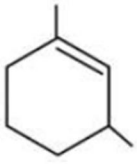
(A) c and d only (B) a and b only

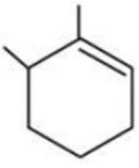
(C) b, c and d only (D) b and c only

40. Which compound would give 3-methyl-6- oxoheptanal upon ozonolysis?

(A) 

(B) 

(C) 

(D) 

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41. In Which of the following reactions, alkanes containing the same number of carbons as the substrate are obtained?

(a) $CH_3CH_2COOH \xrightarrow[\Delta]{\text{soda lime}}$ (b) $CH_3COOK \xrightarrow{\text{electrolysis}}$

(c) $(CH_3)_2CHCOOH \xrightarrow[\text{reducation}]{HI/P}$ (d) $CH_3CH_2Br \xrightarrow[EtOH]{Zn-Cu\ couple}$

(e) $CH_3CH_2Br \xrightarrow[\text{dry ether}]{Na\ and}$

(A) b, c, d (B) c, d, e (C) a, d, e (D) a, e, c

42. Which of the following sodium compound is/are formed when an organic compound containing both nitrogen and sulphur is fused with less amount of sodium?

(A) Sulphite and cyanide (B) Thiocyanate

(C) Cyanide and sulphide (D) Nitrate and sulphide

43. Match List-I with List-II

	List-I (Purification technique)		**List-II (Mixture of organic compounds)**
(A)	Distillation (simple)	(I)	Diesel + Petrol
(B)	Fractional distillation	(II)	Aniline + Water
(C)	Distillation under reduced pressure	(III)	Chloroform + Aniline
(D)	Steam distillation	(IV)	Glycerol + Spent-lye

Choose the correct answer from the options given below :

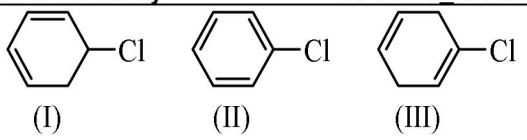
(A) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(B) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(C) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(D) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

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44. 

(I) (II) (III)

Which one is most easily dehydrohalogenated ?

(A) I (B) II (C) III (D) Both A and B

45. Select the most reactive alkene for hydration.

(A)  (B)  (C)  (D) 

SECTION – II
(Numerical Value Answer Type)

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46.

Reaction	Energy Change (in kJ)
$Li\ (s) \rightarrow Li\ (g)$	161
$Li\ (g) \rightarrow Li^{+}(g)$	520
$\frac{1}{2}F_2(g) \rightarrow F(g)$	77
$F(g) + e^{-} \rightarrow F^{-}(g)$	(Electron gain enthalpy)
$Li^{+}(g) + F^{-}(g) \rightarrow LiF(s)$	–1047
$Li(s) + \frac{1}{2}F_2(g) \rightarrow LiF(s)$	–617

Based on data provided, the magnitude of electron gain enthalpy of fluorine would be :

47. XeO_4 molecule is tetrahedral having how many $p\pi - d\pi$ bonds :

48. The density of 3M solution of sodium chloride is $1.0755\,g\ mL^{-1}$. The molality of the solution will be x. Report the answer as 9x :

(molar mass, $NaCl = 58.5\ g\ mol^{-1}$)

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49. In reaction $A+2B \rightleftharpoons 2C+D$, initial concentration of B was 1.5 times of [A], but a equilibrium the concentration of A and B became equal. The equilibrium constant for the reaction is :

50. If x = velocity of election in second Bohr orbit of hydrogen atom and y is the velocity of electron in sixth orbit of He^+ ion. Find $\frac{2x}{y}$.

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If the system of linear equations $3x+y+\beta z=3, 2x+\alpha y-z=-3, x+2y+z=4$ has infinitely many solutions, then the value of $22\beta-9\alpha$ is :

(A) 49 (B) 31 (C) 43 (D) 37

52. If $M(x_0, y_0)$ is the point on the curve $3x^2-4y^2=72$ which is nearest to the line $3x+2y+1=0$ then the value of (x_0+y_0) is equal to

(A) 3 (B) -3 (C) 9 (D) -9

53. Let $f(x)=\left[\frac{\sin x}{x}\right]+\left[\frac{2\sin 2x}{x}\right]+....+\left[\frac{10\sin 10x}{x}\right]$ where [y] is the greatest integer function. The value of $\lim\limits_{x\to 0} f(x)$ is equal to

(A) 55 (B) 164 (C) 165 (D) 375

54. If $f(x)$ is differentiable function with $f'(0)=1$ and

$\cos 2x \sin 2y(f(3x+3y)+f(3x-3y)) = \sin 2x \cos 2y(f(3x+3y)-f(3x-3y))$

for all $x, y \in R$, then the equation of tangent to $y=f(x)$ at $x=\frac{\pi}{2}$ is

(A) $2x+4y=3\sqrt{3}-\pi$ (B) $2x-4y=\pi-3\sqrt{3}$

(C) $2x-y=3\sqrt{3}-\pi$ (D) None of these

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55. If $x_1, x_2, x_3 ..... x_n$ are in A.P. whose common difference is α, then the value of $\sin\alpha(\sec x_1 \sec x_2 + \sec x_2 \sec x_3 + ... + \sec x_{n-1} \sec x_n)$ is

(A) $\frac{\sin(n-1)\alpha}{\cos x_1 \cos x_n}$ (B) $\frac{\sin n\alpha}{\cos x_1 \cos x_n}$

(C) $\sin(n-1)\alpha \cos x_1 \cos x_n$ (D) $\sin n\alpha \cos x_1 \cos x_n$

56. $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \min\{f(t) : 0 \le t \le x\}; & 0 \le x \le 1 \\ 3 - x; & 1 < x \le 2 \end{cases}$ then $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has the value equal to

(A) $\frac{7}{4}$ (B) $\frac{9}{4}$ (C) $\frac{13}{4}$ (D) $\frac{5}{2}$

57. Let $\vec{a} = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}$ & $\vec{c} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$. If V_1 is the volume of parallelopiped whose three coterminous edges are the vectors $\vec{a}+\vec{b}, \vec{b}+\vec{c}, \vec{c}+\vec{a}$ & V_2 is the volume of tetrahedron whose coterminous edges are the vectors $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$, then the value of $(V_1 + V_2)$ is

(A) 256 (B) 278 (C) 288 (D) 289

58. If P and Q are symmetric and non-singular matrices such that PQ = QP, then which of the following is not true?

(A) $QP^{-1} = P^{-1}Q$ (B) PQ is not symmetric

(C) $P^{-1}Q$ is symmetric (D) P^2Q is symmetric

59. If n be a positive integer such that $\sin\left(\frac{\pi}{2n}\right) + \cos\left(\frac{\pi}{2n}\right) = \frac{\sqrt{n}}{2}$, then

(A) $2 \le n \le 6$ (B) $2 < n \le 5$ (C) $8 \le n \le 10$ (D) $4 < n < 8$

60. Let $f(x) = \tan x$ and $g(f(x)) = f\left(x - \frac{\pi}{4}\right)$, where $f(x)$ and $g(x)$ are real valued functions for all possible value of x, then $f(g(x))$ is

(A) $\tan\left(\frac{x-1}{x+1}\right)$ (B) $\tan(x-1) - \tan(x+1)$

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(C) $\frac{f(x)+1}{f(x)-1}$ (D) $\frac{x-\frac{\pi}{4}}{x+\frac{\pi}{4}}$

61. Consider $f(x)=\frac{(x-\alpha)(x-\beta)}{(x-\gamma)(x-\delta)}$ where $0<\alpha<\beta<\gamma<\delta$, then number of local maxima of $f(x)$ will be

(A) 0 (B) 1

(C) 2 (D) Can't be determined

62. A ray of light passing through the point $P(1,2)$ is reflected at a point Q on the x-axis and then passes through (5, 3), then the equation of reflected ray is

(A) $5x-4y=13$ (B) $5x-4y=-3$ (C) $4x+5y=14$ (D) $4x-5y=-6$

63. The value of b, such that the equaton $\frac{b\cos x}{2\cos 2x-1}=\frac{b+\sin x}{(\cos^2 x-3\sin^2 x)\tan x}$ possess solutions is____

(A) $\left(-\infty,\frac{1}{2}\right)$ (B) $\left(\frac{1}{2},\infty\right)$

(C) $(-\infty,\infty)$ (D) $\left(-\infty,\frac{1}{2}\right)\cup(1,\infty)$

64. If $f(x)=x+1, g^{-1}(x)=x^3+x+1$ and $g(f(x))=h(x)$, then $\lim\limits_{x\to 1}\frac{h^{-1}(x)-h^{-1}(1)}{x-1}$ is equal to

(A) 2 (B) 4 (C) 1 (D) 8

65. Let $x=2$ be a root of the equation $x^2+px+q=0$.

and $f(x)=\begin{cases}\frac{1-\cos\left(x^2-4px+q^2+8q+16\right)}{(x-2p)^4}, & x\neq 2p\\ 0, & x=2p\end{cases}$

Then $\lim\limits_{x\to 2p^+}[f(x)]$ is....

Where [.] denotes greatest integer function,

(A) 0 (B) 1 (C) 2 (D) 3

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66. Let the shortest distance between the lines $\frac{x-3}{3}=\frac{y-\alpha}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-\beta}{4}$ be $3\sqrt{30}$. Then the positive value of $5\alpha+\beta$ is

(A) 42 (B) 46 (C) 48 (D) 40

67. If P is a 3×3 matrix such that $P^T=2P+I$, where P^T is the transpose of P and I is the 3×3 identity matrix, then there exists a column matrix $X=\begin{bmatrix}x\\y\\z\end{bmatrix}\neq\begin{bmatrix}0\\0\\0\end{bmatrix}$ such that

(A) $PX=\begin{bmatrix}0\\0\\0\end{bmatrix}$ (B) $PX=X$ (C) $PX=2X$ (D) $PX=-X$

68. A rectangle PQRS has its side PQ parallel to the line $y=5x$ and the vertices P, Q and S on the lines $y=2$, $x=3$ and $x=-3$ respectively. Locus of the vertex R is

(A) $5x-24y+10=0$ (B) $5x-24y+85=0$

(C) $24x-5y+88=0$ (D) $24x-5y+85=0$

69. If $f:R\to R$ be a function such that $f(x)=\begin{cases}x|x|-9; & x\in Q\\ x|x|-\sqrt{2}; & x\notin Q\end{cases}$, then $f(x)$ is

(A) One-one, onto (B) Many one, onto

(C) One-one, into (D) Many one, into

70. The value of product

$$\frac{\sin\frac{\theta}{2}}{\tan^2\frac{\theta}{2}\tan\theta+\tan\frac{\theta}{2}}\times\frac{\sin\frac{\theta}{2^3}}{\tan^2\frac{\theta}{2^3}\tan\frac{\theta}{2^2}+\tan\frac{\theta}{2^3}}\times\ldots\ldots\frac{\sin\left(\frac{\theta}{2^{2n-1}}\right)}{\tan^2\frac{\theta}{2^{2n-1}}\tan\frac{\theta}{2^{2n-2}}+\tan\frac{\theta}{2^{2n-1}}}\times\ldots\ldots$$

upto ∞ is

(A) 1 for $\theta=\frac{\pi}{2}$ (B) 1 for $\theta=\frac{\pi}{4}$

(C) $\frac{1}{2\pi}$ for $\theta=\frac{\pi}{4}$ (D) $\frac{2}{\pi}$ for $\theta=\frac{\pi}{4}$

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. Let $\vec{a}=\hat{i}+2\hat{j}+3\hat{k},\vec{b}=2\hat{i}+3\hat{j}-5\hat{k}$ and $\vec{c}=3\hat{i}-\hat{j}+\lambda\hat{k}$ be 3 vectors. Let $\vec{r}$ be a unit vector along $\vec{b}+\vec{c}$ .If $\vec{r}\cdot\vec{a}=3$ then $6\lambda=$

72. Number of solution of the equation $\sin^4 x-\cos^2 x\sin x+2\sin^2 x+\sin x=0$ in $0\le x\le 3\pi$ is _____

73. If $'A'$ is a square matrix of order less than 4 such that $\left|adj(A)-adj\left(A^T\right)\right|\neq 0$ and $B=4adj(A)$, and if $|A|=4$, then $\left|adj\left(AB^2A^{-1}B\right)\right|+\left|AB^2-B^2A\right|+\left|adj\left(adj(B)\right)\right|=\lambda$, then units digit of λ is_____

74. If principle argument of a complex number lies in $[-\pi,\pi)$, the principal argument of the complex number $"2\sin\frac{6\pi}{5}+i\left(2+2\cos\frac{6\pi}{5}\right)"$ is $"\frac{\lambda\pi}{10}"$ then $|\lambda|=$

75. If $\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{\sqrt{2n+2}-\sqrt{2n-2}}{2+\sqrt{n^2-1}}\right)=\frac{\pi}{2}+\tan^{-1}k$ then find the value of k^2

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